

EE 309: Power Electronics

Course Code : EE 309

Course Name : Power Electronics

L-T-P-C : EE309: 2.5-0.5-0-3

Pre-requisites : IC-160 & IC-160P

Distribution : Elective

Approval: 5th Senate

Course Contents

- **Introduction:** About power electronics; power control through switching; overview of power devices, converters and applications. [2 Lectures]
- **DC to DC power converters:** Buck, boost and buck-boost converters; continuous and discontinuous modes of operation; operation as single-quadrant, two-quadrant and four-quadrant choppers; basics of transformer isolated half-bridge and full-bridge converters. [8 Lectures]
- **Power semiconductor switches:** Desirable switch characteristics; overview of available switches; power diodes / thyristors including GTOs; power MOSFETs / IGBTs; emerging power devices. [8 Lectures]
- **DC to AC inverters:** Background; single phase voltage source inverter; square wave and single pulse PWM (Pulse Width Modulation) operation; selective harmonic elimination; three-phase six-step inverter; simple sine-PWM technique; waveform distortion and harmonics; output filter. [8 Lectures]
- **Engineering aspects:** Drive circuits for power devices; conduction and switching loss calculations; heat sink for power devices; elementary design of magnetic components; basic snubber circuits. [6 Lectures]
- **AC to DC Converters Uncontrolled rectifiers:** Single phase bridge rectifier; centre-tapped rectifier; three phase bridge rectifier. Input side and output side performance; concept of AC input power factor under distorted waveforms. Controlled rectifiers: Some basic concepts regarding AC-DC phase controlled thyristor converters. AC to DC PWM rectifiers: Operation of a PWM bridge DC-AC inverter as an AC to DC PWM rectifier; application in renewable energy systems. [8 Lectures]

Textbooks:

1. Mohan, Undeland and Robbins, **Power Electronics: Converters, Applications and Design**, Wiley India, 2007.
2. L. Umanand, **Power Electronics: Essentials & Applications**, Wiley India, 2009.